

UMAT-OTI workbench

Generate and verify derivatives of an Abaqus UMAT, without editing the subroutine.

1. Source
2. Material model
3. Derivatives and loading
4. Run and results

m3_j2 · fixed form · 1 routine · 0 local Newton solves · NTENS 6 · NSTATV 1 · 4 constants

3. Derivatives and loading

Choose what to compute, and how to load it.

What should be computed?

- ☐ Consistent tangent · DDSdde ?
- ☒ Internal Jacobian of the local solve · FJAC / DF ?
- ☐ Higher-order stress derivatives · orders 2 and above ?
- ☒ Stress sensitivity to material parameters · DSIGMA_DP ?
- ☒ State sensitivity to material parameters · DSTATEV_DP ?

How should the material be loaded?

Loading history ?

Uniaxial strain, 20 steps of 1e-4 (small strain) ▼

Stretches along one axis in 20 equal steps. For a steel-like model this crosses the yield point part way through, so both the elastic and the inelastic response are exercised.

20 increments · provenance: parameter_sensitivity/loading_paths.json default

Before you run

UNSUPPORTED Internal Jacobian of the local solve — This source integrates its law without a local Newton iteration, so it has no internal Jacobian to extract.

The rest of the request can still run: Stress sensitivity to material parameters, State sensitivity to material parameters